

```
In [1]: import pandas as pd
```

```
In [2]: df = pd.read_csv("students.csv")
```

```
In [3]: df
```

```
Out[3]:
```

	Math_Score	Reading_Score	Writing_Score	Placement_Score	Club_Join	Placement_Offer
0	71.0	81.0	65.0	94.0	2024.0	3.0
1	66.0	91.0	63.0	92.0	2024.0	3.0
2	68.0	77.0	76.0	90.0	2022.0	3.0
3	67.0	81.0	78.0	95.0	2020.0	3.0
4	80.0	84.0	78.0	84.0	2023.0	2.0
...	...	...	...	...	...	...
144	77.0	86.0	78.0	77.0	2022.0	2.0
145	0.0	82.0	73.0	93.0	2020.0	10.0
146	77.0	89.0	62.0	93.0	2024.0	10.0
147	63.0	80.0	68.0	90.0	2024.0	3.0
148	71.0	77.0	72.0	93.0	2022.0	3.0

149 rows × 6 columns

```
In [4]: df.info()
```

```

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 149 entries, 0 to 148
Data columns (total 6 columns):
#   Column              Non-Null Count  Dtype  
---  -
0   Math_Score          143 non-null   float64
1   Reading_Score       143 non-null   float64
2   Writing_Score       144 non-null   float64
3   Placement_Score     139 non-null   float64
4   Club_Join           142 non-null   float64
5   Placement_Offer     144 non-null   float64
dtypes: float64(6)
memory usage: 7.1 KB

```

In [5]: `df.describe()`

Out[5]:

	Math_Score	Reading_Score	Writing_Score	Placement_Score	Club_Join	Placement_Offer
count	143.000000	143.000000	144.000000	139.000000	142.000000	144.000000
mean	69.909091	85.062937	70.131944	85.028777	2022.176056	2.569444
std	13.840087	5.953123	6.232384	5.779703	1.400604	1.068643
min	0.000000	75.000000	60.000000	75.000000	2020.000000	1.000000
25%	64.000000	80.000000	65.000000	80.000000	2021.000000	2.000000
50%	70.000000	86.000000	70.000000	86.000000	2022.000000	3.000000
75%	75.000000	90.000000	76.000000	90.000000	2023.000000	3.000000
max	200.000000	95.000000	80.000000	95.000000	2024.000000	10.000000

In [6]: `df.isnull().sum()`

Out[6]:

```

Math_Score      6
Reading_Score   6
Writing_Score    5
Placement_Score 10
Club_Join        7
Placement_Offer  5
dtype: int64

```

```
In [7]: x= df.Math_Score.mean()
```

```
In [8]: df.Math_Score.fillna(x,inplace= True)
```

```
In [9]: df.isnull().sum()
```

```
Out[9]: Math_Score      0
Reading_Score    6
Writing_Score     5
Placement_Score  10
Club_Join        7
Placement_Offer   5
dtype: int64
```

```
In [10]: df.columns
```

```
Out[10]: Index(['Math_Score', 'Reading_Score', 'Writing_Score', 'Placement_Score',
               'Club_Join', 'Placement_Offer'],
              dtype='object')
```

```
In [11]: y = df.Writing_Score.mean()
```

```
In [12]: y
```

```
Out[12]: 70.13194444444444
```

```
In [13]: df.Writing_Score.fillna(y,inplace= True)
```

```
In [14]: df.isnull().sum()
```

```
Out[14]: Math_Score      0
Reading_Score    6
Writing_Score     0
Placement_Score  10
Club_Join        7
Placement_Offer   5
dtype: int64
```

```
In [15]: z = df.Reading_Score.min()
```

```
In [16]: df.Reading_Score.fillna(z,inplace= True)
```

```
In [17]: df.isnull().sum()
```

```
Out[17]: Math_Score      0
Reading_Score    0
Writing_Score    0
Placement_Score  10
Club_Join        7
Placement_Offer   5
dtype: int64
```

```
In [18]: a = df.Placement_Score.max()
```

```
In [19]: df.Placement_Score.fillna(a,inplace= True)
```

```
In [20]: df.isnull().sum()
```

```
Out[20]: Math_Score      0
Reading_Score    0
Writing_Score    0
Placement_Score  0
Club_Join        7
Placement_Offer   5
dtype: int64
```

```
In [21]: df.columns
```

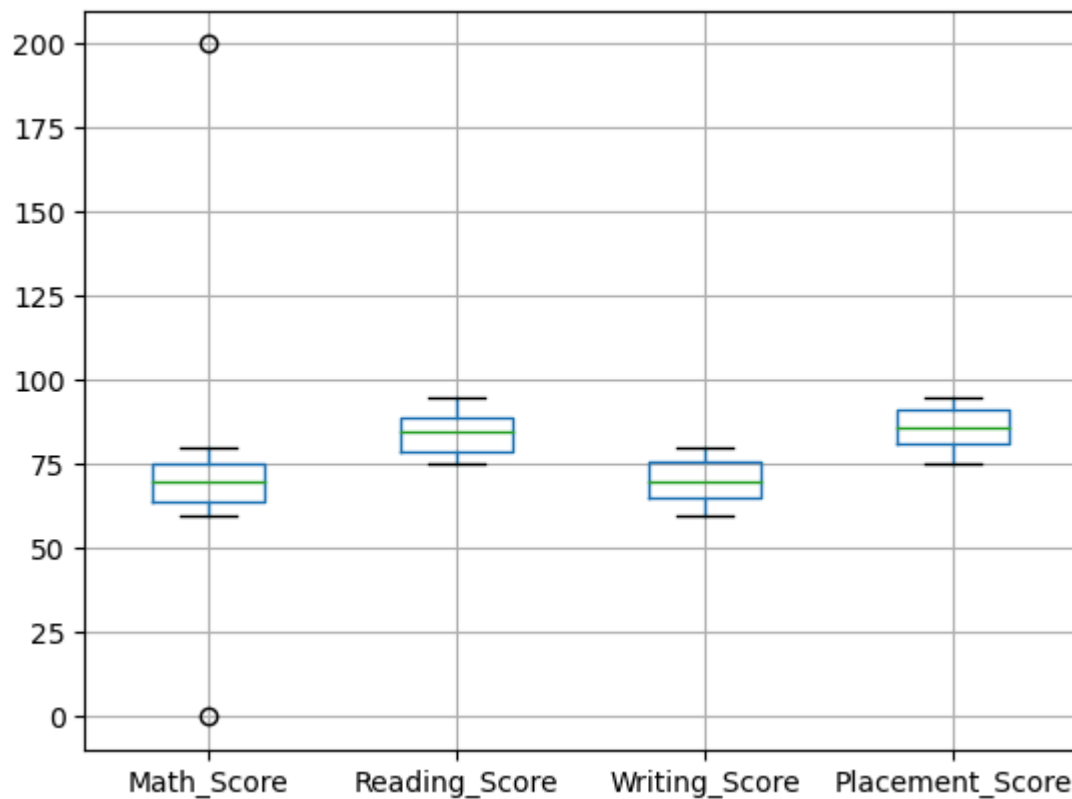
```
Out[21]: Index(['Math_Score', 'Reading_Score', 'Writing_Score', 'Placement_Score',
               'Club_Join', 'Placement_Offer'],
              dtype='object')
```

```
In [22]:
```

```
In [23]: df.boxplot(column=['Math_Score', 'Reading_Score', 'Writing_Score', 'Placement_Score'])
```

Matplotlib is building the font cache; this may take a moment.

```
Out[23]: <Axes: >
```



```
In [26]: import numpy as np
Q1 = np.percentile(df['Math_Score'], 25)
Q3 = np.percentile(df['Math_Score'], 75)

IQR = Q3 - Q1
```

```
In [27]: lower = Q1 - 1.5 * IQR
upper = Q3 + 1.5 * IQR

print("Lower Limit:", lower)
print("Upper Limit:", upper)
```

```
Lower Limit: 47.5
Upper Limit: 91.5
```

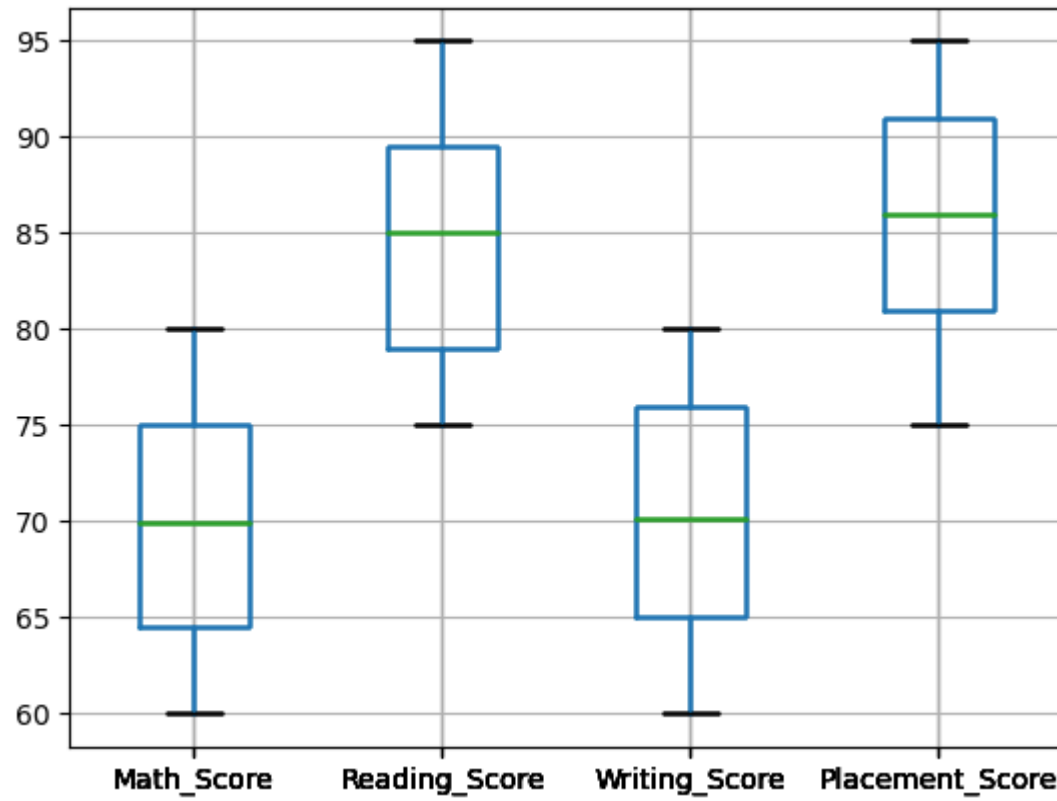
```
In [28]: outliers = df[(df['Math_Score'] < lower) | (df['Math_Score'] > upper)]
print(outliers)
```

	Math_Score	Reading_Score	Writing_Score	Placement_Score	Club_Join	\
141	200.0	89.0	79.0	75.0	2023.0	
145	0.0	82.0	73.0	93.0	2020.0	

	Placement_Offer
141	2.0
145	10.0

```
In [29]: df = df.drop(df[(df['Math_Score'] < lower) |  
                    (df['Math_Score'] > upper)].index)
```

```
In [36]: import matplotlib.pyplot as plt  
  
df.boxplot(column=['Math_Score',  
                  'Reading_Score',  
                  'Writing_Score',  
                  'Placement_Score'])  
  
plt.show()
```



In []: